



The Newtonian Psychologists

THE ACHIEVEMENTS OF THE SCIENTIFIC REVOLUTION REPRESENTED THE vanguard of the **Enlightenment**, that period in European thought in the 17th and 18th centuries when confidence in human reason and experience gradually came to supersede faith in religion and traditional authority. One central feature of Enlightenment thought, which flourished in France, Scotland, England, and Germany, was a commitment to human progress and an optimistic belief in the applicability of scientific knowledge, including social and psychological knowledge, to the improvement of the human condition. The Enlightenment saw the emergence of more liberal, secular, and utilitarian concepts of humanity and the development of more democratic societies, such as the United States. Although not universally embraced, these Enlightenment ideals continue to inform contemporary confidence in the theoretical progress and social utility of the sciences, including social and psychological science.

The rejection of the Aristotelian tradition was good news for the natural sciences. The rejection of Aristotle's geocentric theory and final causal explanations of motion led to advances in astronomy and physics. However, it was not so obviously good news for psychology. One of the casualties of the scientific revolution was Aristotle's biologically grounded functional psychology, which came to be replaced by a variety of mechanistic psychological theories. This was not the intent of the pioneers of the new science, such as Galileo, Bacon, and Newton. Although they maintained that final causal explanation has no place in physical science, most recognized that final causal explanation is entirely appropriate in the realm of human and animal behavior. Yet this qualification was generally ignored by the protopsychologists of the 17th and 18th centuries, who tried to create a science of psychology based upon the mechanistic forms of efficient causal explanation characteristic of the new science, for which Newton's physics came to serve as a paradigm.

Newton's theory of universal gravitation was hugely influential, not only in physical science, where it continued to reign supreme throughout the 18th and 19th centuries, but also with respect to the forms of psychological theory that developed during these centuries. These were either attempts to model psychological theory upon Newtonian science, such as associationist psychology, or

reactions to them, such as “common sense” psychology, and rationalist, humanist, and romantic alternatives. Newton’s theory remained influential in psychology into the 20th century (and remains influential in the 21st), even for those forms of functionalist and behaviorist psychology that were supposedly grounded in Darwin’s evolutionary biology. For example, even though Newton’s theory had been decisively rejected by natural scientists by the early decades of the 20th century, throughout his professional career the behaviorist psychologist Clark C. Hull prominently displayed a copy of Newton’s *Principia* (1687) on his desk and made it required reading for his graduate students in psychology.

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Like many other social scientists, psychologists came to treat Newton’s theory not only as a paradigm of scientific achievement, but also as a paradigm of scientific thought. They came to treat particular features of Newton’s theory as essential conditions of scientific thought, which ought to be reproduced in any properly scientific psychology.

Newtonian Science

One of the central features of Newton’s theory was that it provided a *universal* explanation of terrestrial and celestial motion in terms of gravitational forces. This feature of Newton’s theory became fairly quickly elevated into an implicit criterion of adequacy for a scientific explanation. According to this criterion, an adequate scientific explanation must be universal: It must furnish an explanation of *all* the phenomena in any particular domain. Newton himself gave no more than a cautious and qualified endorsement of this principle, allowing that in at least some cases, we might have to recognize different causal explanations of the same range of phenomena: “Therefore to the same natural effects we must, *as far as possible*, assign the same causes” (1687/1969, p. 398, my emphasis). In his own case, he was prepared to introduce God’s intervention as the cause of some celestial motions, over and above gravitational forces, and thought it entirely appropriate to do so “to discourse of whom [God] from the appearances of things does certainly belong to natural philosophy” (1687/1969, p. 546). Nonetheless, generations of psychologists since Newton have been committed to the notion that universality is the mark of a genuine scientific explanation (Kimble, 1995).

The objects of Newton’s theory were material bodies, with mass and velocity, held to exist for all time and in all regions of space. Later generations of psychologists likewise presumed that human psychology and behavior are invariant in historical time and cultural space, to the point of claiming that it is unscientific to suppose that this might not be the case (Spence, 1987).

Two other principles that figured prominently in Newton's science were successfully adopted by later generations of physical scientists. These were the principle of *atomism*, which holds that entities that form the subject matter of scientific disciplines, such as atoms, planets, and cells, can be individuated without reference to other entities and can exist independently of them, and the principle of *explanatory reduction*, which holds that the best explanation of a complex entity is in terms of its material components. At the beginning of *Principia*, Newton speculated that the properties and behavior of complex material bodies might eventually be explained in terms of mathematical laws governing the material "corpuscles" (or atoms) that composed them, the **corpuscularian hypothesis** promoted by his colleague Robert Boyle:

I wish we could derive the rest of the phenomena of nature by the same kind of reasoning from mathematical principles; for I am induced by many reasons to suspect that they may all depend upon certain forces by which the particles of bodies, by some causes hitherto unknown, are either mutually impelled toward each other, and cohere in regular figures, or are repelled and recede from each other.

—(1687/1969, p. xvii)

This hypothesis, developed with great success in the physical sciences, led many psychologists to presume that an atomistic and reductive analysis of human psychology and behavior is also required to constitute psychology as a genuine science.

Newton's gravitational theory is often treated as a paradigm of a deterministic physical system: a system in which, for every event, there is an antecedent set of conditions sufficient to produce it. However, it was the French mathematician Pierre Simon de Laplace (1749–1827) who gave Newton's theory its deterministic interpretation, by demonstrating the stability of the solar system within Newtonian mechanics. Newton had postulated the intervention of God to maintain this stability, but Laplace famously remarked to Napoleon that he had no need of that hypothesis.

Laplace represented his deterministic thesis through the image of a superintelligence capable of knowing the position and velocity of all material bodies at any instant in time and thus able to calculate and successfully predict all consequent positions and motions according to Newton's laws. Later generations of psychologists came to treat the principle of determinism as a foundational assumption of scientific psychology, despite the fact that physicists abandoned it in the early part of the 20th century.

The success of Newtonian science also promoted a plausible legend, that progress in science is achieved through a process of continuous **theoretical unification**. Thus Galileo's law of free fall and Kepler's laws of planetary motion were independently developed and restricted to terrestrial and celestial motion respectively, but

were later unified by integration within Newton's gravitational theory. The various gas laws relating pressure, temperature, and volume, Graham's law, Charles's law, and Boyle's law, were independently established by Thomas Graham (1805–1869), Jacques Alexandre Charles (1746–1823), and Robert Boyle but were later integrated within the kinetic theory of gases; and contemporary physicists look forward to the attainment of Grand Unified Theory (GUT), which they hope will eventually integrate the theories of the four known physical forces (gravitation, electromagnetism, strong nuclear, and weak nuclear). Similarly, psychologists have tried to develop theories of learning that integrate classical and operant conditioning (e.g., Hull, 1937) and regularly dream of developing a universal theory that will integrate the various explanatory principles of the different branches of psychology. Yet, as is often lamented, psychology still awaits its Newton.

Newton's achievement is often also associated with an **empiricist or positivist conception of scientific explanation**, according to which causal explanations and scientific laws are nothing more than descriptions of observational correlation, with references to causal forces and hypothetical entities dismissed as appeals to "occult" properties. At the end of *Principia*, Newton admitted that he did not know the nature or ultimate cause of gravity, and declared, "I frame no hypothesis" (1687/1969, p. 547). Yet Newton was no empiricist or positivist. He certainly conceived of gravity as a genuine force and was not averse in principle to postulating hypothetical entities to furnish causal explanations: For example, in *Opticks* (1704/1952) he postulated "multitudes of unimaginable small and swift corpuscles" to explain the transmission of light. Later 18th-century empiricist interpreters, such as George Berkeley and David Hume, and 19th-century positivists, such as Auguste Comte (1798–1857) and Ernst Mach (1838–1916), were the ones who characterized Newton's achievement in these terms and who mediated the adoption of the Newtonian paradigm for many psychologists.

Newton was, of course, an empiricist in the general sense that any post-Galilean scientist was a (methodological) empiricist. He maintained that scientific theories must be based upon empirical data rather than rational intuition or classical authority:

Although the arguing from experiments and observations by induction be no demonstration of general conclusions, yet it is the best way of arguing which the nature of things admits of.

—(1704/1952, p. 404)

Yet Newton did not claim that scientific explanations are nothing more than descriptions of observational correlation. He did not presume that all properly scientific explanations are universal and reductive or that the objects of all scientific disciplines are atomistic and invariant in space and time. And he never denied the legitimacy of final causal explanations of human behavior.

Yet all this was ignored by later generations of psychologists who tried to reprise Newton's achievement by reproducing these particular features of Newtonian theory. They strove to develop universal theories of mentality and behavior, conceived as atomistic entities invariant in (cultural) space and (historical) time. They sought to establish mechanistic laws describing the combination of mental elements into mental complexes and the correlation of mental states with other mental states and behavior, and they hoped to eventually explain them in terms of underlying laws of neurophysiological combination and correlation.

John Locke: The Underlaborer for Newtonian Science

The English philosopher John Locke (1632–1704) was the first to systematically apply the principles of Newtonian science to psychology and is generally credited as the father of British empiricism. Locke was greatly impressed by the “incomparable Mr. Newton” and his scientific achievements. They met in 1689 and remained friends and correspondents throughout the rest of their lives. Locke avowed that his intellectual ambition was to serve as a kind of “under-labourer” for Newtonian science, “clearing ground a little, and removing some of the rubbish, that lies in the way to knowledge” (1690/1975, p. 10).

Locke was born in Somerset, England, and educated at Oxford University, where he studied medicine and attained teaching appointments in Greek, moral philosophy, and rhetoric. He demonstrated his medical skill in 1668 when he supervised an operation to remove a hydatid cyst of the liver from the first Earl of Shaftesbury, who became his friend and patron (the silver tap inserted in Shaftesbury's liver made him the object of many contemporary witticisms).

While studying at Oxford, Locke met Robert Boyle, the pioneer of modern chemistry and primary advocate of the corpuscularian hypothesis. Locke served as Boyle's research assistant for some years, and both became members of the newly founded Royal Society (Boyle was one of its founders). Locke lived in France from 1675 to 1679, where he read the works of Descartes and Gassendi.

Like Hobbes before him, Locke was greatly influenced by the political upheavals of his day. He may have seen Charles I executed in the courtyard of the Palace of Whitehall. This is quite likely, since at the time the 17-year-old Locke was attending Westminster School, which borders on the Palace of Whitehall, and it was customary in his day for schoolchildren to be taken on outings to witness public executions (they were usually beaten afterward, to impress the event on their memory). Like his mentor Lord Shaftesbury, Locke was opposed to the Catholic Stuart kings of England and in particular to their defense of the doctrine of the “divine right” of kings. Locke claimed that political sovereignty depends upon the consent of the governed. He maintained that the power of a state

should be constrained by a system of checks and balances, particularly between the executive, legislative, and judicial branches of government. His defense of the fundamental rights of citizens in a democratic state, developed in his *Two Treatises on Government* (1689/1988), played an influential role in the development of the United States Constitution and the state constitutions of Virginia and North Carolina.

Shaftesbury's opposition to the succession of the Catholic James II led to his imprisonment in the Tower of London. Fortunately, he escaped, and sought exile in Holland in 1681. Locke followed him into exile in 1683 but returned to England in 1688 after the accession of the Protestant William of Orange to the English throne (the "Glorious Revolution"). Two years later Locke published his major work, *An Essay Concerning Human Understanding* (1690), which he revised through five subsequent editions (the last in 1706, published posthumously). On his return to England, Locke served in a number of minor administrative positions in government. He spent his last years (1691–1704) in Essex at the home of Lady Masham (1658–1708), an early proponent of women's education.

Psychological and Meaning Empiricism Locke's aim in the *Essay* was to determine the possibility and extent of human knowledge by exploring the origin of our ideas. Like Hobbes, he was a psychological empiricist, who held that all our ideas or concepts are derived from experience, either from sense experience, in the case of "ideas of sensation," or from inner experience, in the case of "ideas of reflection" (1690, pp. 104–105). He was also an **epistemological empiricist**, who maintained that all knowledge is derived from experience. This was a natural consequence of his psychological empiricism, since knowledge for Locke amounted to "nothing but the perception of the connection and agreement, or disagreement and repugnancy of any of our ideas" (1690/1975, p. 525).

Locke was a **psychological atomist**, who claimed that mental states and properties could be individuated independently of each other. He maintained that the basic materials of our psychology and knowledge are simple ideas of color, taste, smell, and the like, which are "perfectly distinct" from each other (1690/1975, p. 119). For Locke, simple ideas of sensation constitute the atoms or corpuscles of our complex ideas and knowledge of the external world. From these simple ideas, we can form an "almost infinite variety" of complex ideas, such as the ideas of "material substance," "identity," "infinity," and the like, via the mental operations of comparison, memory, discrimination, combination, enlargement, abstraction, and reasoning. All ideas are simple ideas or composed of simple ideas, just as material bodies are either elementary corpuscles or composed of elementary corpuscles. Thus our complex ideas of material substances such as apples and antelopes are "nothing but collections of simple *ideas*, with a supposition of something, to which they belong, and in which they subsist" (1690/1975, p. 316).

Locke maintained that it is not possible to have any simple idea that is not derived from experience or to invent any complex idea that is not constructed out of simple ideas derived from experience. Consequently he claimed that a congenitally blind person could not have any idea of a color such as red, and anyone who lacked experience of color, taste, and smell could not form the complex idea of a material substance such as an apple.

Locke was also a meaning empiricist, who claimed that words derive their meaning through their employment as “signs” for ideas derived from experience:

Man, therefore, had by nature his organs so fashioned, as to be fit to frame articulate sounds, which we call words. But this was not enough to produce language; for parrots . . . will be taught to make articulate sounds distinct enough, which yet, by no means, are capable of language.

Besides articulate sounds, therefore, it was further necessary that he should be *able to use these sounds, as signs for internal conceptions*; and to make them stand as marks for the *ideas* within his own mind, whereby they might be made known to others, and the thoughts of men’s minds be conveyed from one to another.

—(1690/1975, p. 402)

Consequently Locke maintained that the meaningful use of language is also dependent upon experience. A congenitally blind person could not understand the meaning of the word *red*, and anyone who lacked experience of color, taste, and smell could not understand the meaning of the word *apple*.

Primary and Secondary Qualities Locke’s psychological theory was not only modeled upon Newton’s theory, but also was employed to articulate and justify the central tenets of the Newtonian worldview. Thus Locke endorsed the ancient distinction between primary and secondary qualities championed by Newton and earlier pioneers of the scientific revolution such as Descartes and Galileo (although the terms of the distinction were coined by Boyle). Locke agreed that material substances have only primary qualities such as shape, size, and motion and that secondary qualities such as colors and tastes are merely the effects of our sensory interactions with material substances with primary qualities:

The particular bulk, number, figure, and motion of the parts of fire, or snow, are really in them, whether any ones senses perceive them or no: and therefore they may be called real qualities, because they really exist in these bodies. But light, heat, whiteness, or coldness, are no more really in them, than sickness or pain is in manna [a form of bread].

—(1690/1975, pp. 137–138)

However, Locke drew the distinction in a slightly different way from Newton and earlier scientists. He distinguished primary and secondary qualities in terms

of the different powers of material substances to cause *ideas* of primary and secondary qualities. Locke claimed that although material substances really do have primary qualities that form the basis of their causal powers, they have secondary qualities only in the sense that they have the power to cause ideas of secondary or “sensible” qualities in us.

Since Locke believed that material substances really do have primary qualities but not secondary qualities, he claimed that only our ideas of primary qualities could be said to *resemble* the primary qualities of material substances themselves:

From whence I think, it is easy to draw this observation, that the *ideas of primary qualities* of bodies, are *resemblances* of them, and their patterns do really exist in the bodies themselves; but the *ideas, produced in us by these secondary qualities, have no resemblance* of them at all. There is nothing like our ideas, existing in the bodies themselves. They are, in the bodies, we denominate from them, only a power to produce those sensations in us, and what is sweet, blue, or warm in idea, is but a certain bulk, figure, and motion of the insensible parts, in the bodies themselves, which we call so.

—(1690/1975, p. 137)

The justification of this claim was another matter. Like Descartes and Hobbes, Locke claimed our ideas are like pictures, or images derived from sense experience:

The *ideas* of the nurse and the mother, are well framed in their minds [children’s]; and, like pictures of them there, represent only those individuals.

—(1690/1975, p. 411)

Although it was reasonable to suppose that our ideas or concepts are images that resemble the contents of our sense perception (if they are simply weaker versions of them), it was much more problematic to suppose that our ideas are images that resemble the (primary) qualities of material substances in the external world, since we have no means of comparing our imagistic ideas of objects with the objects themselves (for example, our idea of the rectangular shape of a table with the shape of the table itself). Indeed, on this conception, it was very hard to justify the assumption that we could have knowledge of the material substances that formed the basic substratum of the Newtonian universe. Locke himself admitted that “of this supposed something, we have no clear distinct *idea* at all” (1690/1975, p. 316).

Consciousness Like Descartes, Locke held that self-knowledge of mental states, including our sense experience, is direct and certain, because we are immediately

conscious of our mental states. He claimed that consciousness accompanies all mental states and makes our psychology transparent to us:

Consciousness . . . is inseparable from thinking, and as it seems to me, essential to it; it being impossible for anyone to perceive, without perceiving, that he does perceive. When we see, hear, smell, taste, feel, meditate, or will anything, we know that we do.

—(1690/1975, p. 335)

Locke's reasons for holding this view may appear rather lame to the modern reader. He claimed that it is "hard to conceive, that any thing should think, and not be conscious of it":

For to be happy or miserable without being conscious of it, seems to me utterly inconsistent and impossible.

—(1690/1975, p. 110)

It is perhaps not so hard to conceive of this at all, and later theorists who were prepared to recognize unconscious mental states and processes challenged Locke's view. However, this conception of mental states as essentially conscious, and objects of direct and certain knowledge, remained popular among psychologists and philosophers until the early 20th century.

Locke seems to have held this view because he treated consciousness of mental states as a form of internal perception, or introspection. This notion of consciousness as a form of inner awareness, which also proved popular with later generations of psychologists and philosophers, was itself a 17th-century invention. The earliest recorded use of the English verb form *be conscious of* in this sense dates from 1620, and of the noun *consciousness* from 1678. *Self-consciousness* in this sense first appeared in 1690, the year in which Locke's *Essay* was first published. Of course, the term *consciousness* existed long before this, but meant something quite different; its original etymological meaning was "shared knowledge" (*con*, "with," + *scio*, "I know"). So too with the concept of introspection: The term "introspection" (*intro*, "within," + *specio*, "I look") made its first appearance in France and England toward the end of the 17th century (Wilkes, 1988).

Probable Opinion Locke's commitment to direct and certain knowledge of mental states, including the contents of our sensory experience, left him with a problem about our knowledge of material substances in the external world. Our beliefs about such entities appeared to be nothing more than uncertain inferences based upon sensory experience.

However, Locke was quite prepared to embrace this conclusion, and it bothered him far less than it had Descartes. Descartes was vexed by this problem

because he had retained the Aristotelian view that genuine scientific knowledge is necessarily true and demonstrable. However, scientists and philosophers gradually abandoned this view as the scientific revolution developed, for they came to recognize that scientific knowledge is fallible, revisable, and at best merely *probable*. The modern conception of probability was itself a late development of the scientific revolution (Hacking, 1975). Thus Locke, like Newton, was willing to accept that we can have only “probable opinion” about material substances and their properties and indeed with respect to any form of scientific knowledge.

The Association of Ideas Locke’s psychological empiricism inclined him toward a rather naive and optimistic environmentalism, of the sort characteristic of 20th-century behaviorist psychologists such as John B. Watson and B. F. Skinner (1904–1990). Because he believed that most human failings are the product of poor upbringing, Locke stressed the critical importance of a good education in *Some Thoughts Concerning Education* (1693/1989).

It was in the course of considering unreasonable adult behavior and irrational childhood fears that Locke discussed the “association of ideas.” He distinguished between associations based upon natural connections and those based upon contingent or “accidental” contiguity:

Some of our *ideas* have a natural correspondence and connection one with another: It is the office and excellency of our reason to trace these, and hold them together in that union and correspondence which is founded in their particular beings. Besides this there is another connection of *ideas* wholly owing to chance or custom; *ideas* that in themselves are not at all of kin, come to be so united in some men’s minds, that t’is very hard to separate them, they always keep in company, and the one no sooner at any time comes into the understanding but its associate appears with it.

—(1690/1976, p. 395)

Locke claimed that ideas associated by contiguity lead to unreasonable, unnatural, and superstitious beliefs:

Many children, imputing the pain they endured at school to their books they were corrected for, so join these *ideas* together, that a book becomes their aversion, and they are never reconciled to the study and use of them all their lives after; and thus reading becomes a torment to them, which otherwise possibly they might have made the great pleasure of their lives.

—(1690/1976, p. 399)

Given that many “vain terrors” are based upon such contingent associations, Locke reasoned that they could be relieved by attenuating the association. In



Locke's explanation of the creation of "vain terrors" of the dark through "unnatural association" with stories of "Goblins and Sprights" (terrors that later afflicted the behaviorist John B. Watson).

explaining how to relieve a child's unnatural fear of frogs, Locke anticipated the behavioral therapy of "systematic desensitization" developed by Joseph Wolpe in the 1950s (Wolpe, 1969):

Your child shrieks, and runs away at the sight of a frog; let another catch it and lay it down a good distance from him; at first accustom him to look upon it, and see it leap without emotion; then to touch it lightly while it is held fast in another's hand; and so on till he can come to handle it as confidently as a butterfly, or a sparrow. By the same way any other vain terror may be removed if care be taken, that you go not too fast, and push not the child on to a new degree of assurance, till he be thoroughly confirm'd in the former. And thus the young soldier is to be trained on to the warfare of life.

—(1693/1989, p. 151).

Locke is often credited as the founder of the later tradition of associationist psychology, but this is misleading, since he did not believe that the principle of contiguity provided a universal explanation of human thought and behavior. On the contrary, he insisted that it only explained a limited range of "unreason-

ableness" within it. The point of education was not to manipulate thought and behavior on the basis of associations grounded in contiguity, but rather to "prevent the undue connection of ideas in the minds of young people" (1690/1976, p. 397).

Locke's *Essay* had a major impact in Britain and Europe, where it was favorably received. The favorable reception was at least partly due to Locke's careful refusal to speculate about the relation between mental states and material states of the brain. Although Locke entertained the "supposition" that mental states are states of material substance, he maintained that "the more probable opinion" is that they are states of an "individual immaterial substance" (1690/1976, p. 345).

George Berkeley: Idealism

George Berkeley was born in Kilkenny, Ireland, where he first attended college. He entered Trinity College, Dublin in 1700, at the age of 15. He received his bachelor's degree in 1704 and his master's degree in 1707. He became a fellow of the college in 1707 and was later ordained as a deacon of the Anglican Church. In 1709 he published *An Essay Towards a New Theory of Vision*, followed a year later by *A Treatise Concerning the Principles of Human Knowledge* (1710).

After traveling in Europe for some years, in 1724 Berkeley embarked on an ambitious scheme to found a Christian college in Bermuda for the education of both colonists and native peoples. He sailed for America in 1728 with his new wife, settling in Rhode Island to await financial support for his project from the British government. When it became clear that the promised support would not be forthcoming, he returned to London in 1731. He was appointed Bishop of Cloyne (in County Cork, Ireland) in 1734 and devoted the last years of his life to his parishioners, the intended audience of his last work *Siris: A Chain of Philosophical Reflections and Inquiries Concerning the Virtues of Tar-Water* (1744). The city of Berkeley, California, is named after him.

Idealism Berkeley is best remembered as the philosophical advocate of **idealism**: the view that only immaterial minds and their ideas exist. The Irish poet William Butler Yeats enthused that Berkeley “proved the world a dream,” and Dr. Johnson is reputed to have refuted Berkeley by kicking a stone: “I refute him thus!” When he visited the home of Jonathan Swift, Swift declined to open the door, declaring that it was merely an idea in Berkeley’s mind. Berkeley was not moved by such responses but did take precautions to ensure that on his death his body would be laid out until it began to putrefy (he was afraid of being buried alive while in a comatose state).

Whatever the limitations of his idealism, Berkeley did highlight some serious problems with the Newtonian theoretical system. He questioned how primary qualities such as size, shape, and motion could be held to be objective properties of material substances if our only access to these properties is through our sense experience of subjective and variable secondary qualities such as color. He also developed some of the less palatable implications of the Lockean “way of ideas” and drew attention to a problem of perception that became a major focus of psychologists in the 19th and early 20th centuries.

Berkeley agreed with Locke’s definition of knowledge in terms of the “agreement or repugnancy among ideas,” but poured scorn on Locke’s assumption that we can attain knowledge of the primary qualities of material substances through our ideas of them. Since according to Berkeley (and Locke) we have no independent access to material substances, we have no reason whatsoever to suppose that our ideas of primary qualities resemble qualities that material substances really

do have. As Berkeley tersely put it, “an idea can be like nothing but an idea” (1710/1975, p. 79).

Thus Berkeley claimed that our ideas of material substances cannot make reference to entities held to be distinct from and independent of our sensory ideas, since all we have access to are complexes of sensory ideas that exhibit a certain “constancy and coherence” in our experience. Accordingly, our thought and talk about “material substances” such as apples, trees, and tables can only reference such complexes of ideas:

I see this *cherry*, I feel it, I taste it. . . . Take away the sensations of softness, moisture, redness, tartness, and you take away the cherry. Since it is not a being distinct from sensations; a *cherry*, I say, is nothing but a congeries of sensible impressions, or ideas perceived by various senses: which ideas are united into one thing (or have one name given to them) by the mind; because they are observed to attend each other. . . . But if you mean by the word *cherry* an unknown nature, distinct from all these sensible qualities, and by its existence something distinct from its being perceived; then, indeed, I own, neither you nor I, nor anyone else, can be sure it exists.

—(1713/1975, pp. 196–197)

According to Berkeley, it is meaningless to claim that such an object exists, since we have no sense impression or idea of it.

Consequently, Berkeley rejected as absurd the Newtonian program of explaining the perceived properties of material substances in terms of the properties of the corpuscles that compose them. He claimed that the notion that our sense impressions “are the effects of powers resulting from the configuration, number, motion and size of corpuscles, must certainly be false” (1710/1975, p. 84). He also employed the principle of meaning empiricism to reject central components of Newton’s theory. For example, talk of an absolute space that could exist in the absence of material bodies is nonsense, according to Berkeley, since no one could perceive it:

And so let us suppose that all bodies were destroyed and brought to nothing. What is left they call absolute space. . . . that space is infinite, immovable, invisible, insensible, without relation and without distinction. That is, all its attributes are privative or negative. It seems therefore to be mere nothing. . . .

From absolute space then let us take away now the words of the name, and nothing will remain in sense, imagination or intellect. Nothing else then is denoted by these words than pure privation or negation, i.e. mere nothing.

—(1721/1975, p. 222)

For similar reasons, he rejected “gravity” and other mechanical forces as “occult” qualities that explain nothing. According to Berkeley, references to mechanical

efficient causes or “powers” are vacuous, since we have no sense impressions or ideas of them. Thus he maintained that “real efficient causes of the motion and existence of bodies or of corporeal attributes in no way belong to mechanics or experiment, nor throw any light on them” (1721/1975, p. 219).

For Berkeley, the only business of science, including Newtonian science, is to determine “by experiment and reasoning” the regularities to be found in our sensory experience. He denied that efficient causes can be identified in sense experience: The only efficient cause is God, the creator of sense impressions and ideas, who ensures that certain complexes are reliably conjoined in our experience. Scientific laws merely describe conjunctions between “sign” and “signified” that are useful in anticipating experience, and are maintained through the beneficence of God:

The ideas of sense . . . are not excited at random . . . but in a regular train or series, the admirable connection whereof sufficiently testifies the wisdom and benevolence of its Author. Now the set rules or established methods, wherein the mind we depend upon [God] excites in us the ideas of sense, are called the *Laws of Nature*: and these we learn by experience, which teaches us that such and such ideas are attended with such and such other ideas, in the ordinary course of things.

—(1710/1975, pp. 85–86)

Distance Perception Berkeley also engaged a problem about visual perception that vexed later generations of psychologists and physiologists. How do we perceive the distance, shape, size, and motion of physical bodies in space and time? According to Berkeley, the primary objects of perception are simple and discrete (atomistic) sensory impressions or ideas: of color, smell, sound, and the like. Yet, if this is the case, we do not strictly perceive the distance, shape, size, and motion of physical bodies, since we do not have discrete (atomistic) sensory impressions of them, and our ideas or concepts of them cannot be derived from the mere aggregation or association of sensory impressions of color, smell, sound, and the like.

In *An Essay Towards a New Theory of Vision* (1709/1975), Berkeley maintained that distance is not visually perceived:

It is, I think, agreed by all that distance, of itself and immediately, cannot be seen. For distance being a line directed end-wise from the eye, it projects only one point in the fund of the eye, which point remains invariably the same, whether the distance be shorter or longer.

—(1709/1975, p. 9)

Berkeley rejected the account that Descartes and his followers championed. They claimed that the visual perception of distance is a “complicated reasoning

process" involving calculations based upon differences in the angle of a triangle formed by imaginary lines from the eyes to the object perceived. Berkeley dismissed this account because the postulated cognitive process is not an object of experience:

But those lines and angles, by means whereof some men pretend to explain the perception of distance, are themselves not at all perceived, nor are they in truth ever thought of by those unskillful in optics. I appeal to anyone's experience whether upon sight of an object he computes its distance by the bigness of the angle made by the meeting of the two optic axes? Or whether he ever thinks of the greater or lesser divergency of the rays, which arrive from any point to his pupil? Everyone is himself the best judge of what he perceives, and what not. In vain shall any man tell me that I perceive certain lines and angles which introduce into my mind the various ideas of distance, so long as I myself am conscious of no such thing.

—(1709/1975, p. 10)

According to Berkeley, we make inferences about distance on the basis of associations between visual cues, bodily movements, and tactile sensations. We learn that bodies that appear smaller in our visual field and require a broader visual focus are at a greater distance than those that appear larger and require a narrower visual focus, based upon their correlation with past experience of moving through space to locate them:

Looking at an object I perceive a certain visible figure and color, with some degree of faintness and other circumstances, which from what I have formerly observed, determine me to think, that if I advance forward so many paces or miles, I shall be affected with such and such ideas of touch.

—(1709/1975, p. 20)

The inadequacy of Berkeley's account, on its own terms, is worth noting. If the idea or concept of distance cannot be derived from visual sense impressions (since distance cannot be seen), it cannot be derived from tactile sense impressions either (since distance cannot be touched). Berkeley only appeared to avoid the problem by supposing that distance is inferred, but this presupposes that we already have an idea or concept of distance that we employ in our inferential judgment. That is, although a learned correlation between visual and tactile sense impressions may explain our developed ability to estimate distance, it cannot explain the origin of our idea or concept of distance. Such considerations led later theorists such as Thomas Reid (1710–1796) and Immanuel Kant (1724–1804) to claim that our concepts of distance, shape, and size, and our concepts of substance and causality, are part of our innate endowment, and Herman von Helmholtz (1821–1894) to

theorize that our “perception” of physical bodies and their properties is really a form of unconscious cognitive judgment.

David Hume: Mental Mechanism

David Hume (1711–1776) was born in Edinburgh, Scotland, the son of a minor landowner. He studied law at Edinburgh University, but never took a degree. In 1734 he traveled to La Flèche (where Descartes had studied at the Jesuit College), which he made his base for studying and writing. At the age of 28 he produced his major theoretical work, *A Treatise of Human Nature, Being an Attempt to Introduce the Experimental Method of Reasoning into Moral Subjects* (1739). Hume later complained that “it fell still-born from the press, without reaching such distinction as even to excite a murmur from the zealots.” This was somewhat of an exaggeration, but it did not attract as much attention as Hume had wished. This motivated him to produce a shorter work in which he presented the main themes of the *Treatise* in a more accessible manner. In 1748 he published *An Enquiry Concerning Human Understanding*, followed by *An Enquiry Concerning the Principles of Morals* in 1751.

Hume’s skeptical views did not recommend him to the establishment of his day. He was refused professorships at Edinburgh and Glasgow universities because of religious opposition. He returned to Edinburgh in 1739, where he produced *Essays Moral and Political* in 1742 and his *History of England* between 1754 and 1762. He served as an aide to General James St. Clair during the years 1746–1748 and became the darling of the French salons during the years 1763–1766, when he served as secretary to the embassy and later as chargé d’affaires in Paris.

Hume died in Edinburgh in 1776. The Scottish church fathers that had persecuted him throughout his lifetime tried to convert him as he lay in bed dying, but he chased them out. He was interred in a modest tomb in Edinburgh, and it was not until 1997 that a monument was finally erected to celebrate his contribution to the Scottish Enlightenment. Hume’s last skeptical work on religion, *Dialogues Concerning Natural Religion*, which he withheld from publication during his lifetime, was published posthumously in 1779. He entrusted the work to the economist Adam Smith (1723–1790), who declined to publish it (it was eventually published by Hume’s nephew).

Hume was perhaps the most consistent empiricist, pressing the consequences of psychological and meaning empiricism to their limit. Although he ranks as one of the major thinkers of the Western tradition, his recognition had been grudging within the philosophical community, at least until the 20th century. Hume’s critical arguments were invariably destructive and skeptical. His devastating critiques of our pretension to have knowledge of material substances, causality, the future, and the self infuriated later generations of philosophers and scientists, who made valiant attempts to answer them, with doubtful success.

Much of the original antagonism to Hume derived from his atheism, which prevented him from obtaining any university position during his lifetime. However, much of it also derived from Hume's honestly avowed ambition to make a literary name for himself. Philosophers are supposed to be disinterested seekers of truth. It came hard to many that one with such seemingly base and selfish motives could have the best arguments. Hume's literary reputation was established in his own day primarily through his *History of England*, although his *Treatise* and *Enquiries* account for his enduring reputation.

Hume's primary aim in the *Treatise* and *Enquiries* was to provide naturalistic psychological explanations of how we come to hold those beliefs about material bodies, causality, and the self that he maintained are incapable of rational justification. In advancing these psychological explanations, Hume initiated the tradition of associationist psychology and developed a hugely influential account of causal explanation grounded in the principles of association.

Indeed, Hume saw himself primarily as a psychologist applying the principles of Newtonian science to the study of the human mind. He subtitled the *Treatise* as *An Attempt to Introduce the Method of Experimental Reasoning Into Moral Subjects*, and his investigations were very much in the spirit of the new science of Galileo and Newton. He never accepted any claim about human psychology at face value but always checked it for himself introspectively. For example, he rejected the claim that we have direct knowledge of a simple self by reporting his failed attempt to identify it through introspection:

There are some philosophers, who imagine we are every moment intimately conscious of what we call our *self*. . . .

For my part, when I enter most intimately into what I call *myself*, I always stumble upon some particular perception or another, of heat or cold, light or shade, love or hatred, pleasure or pain. I never can catch *myself* at any time without a perception, and never can observe anything but the perception.

—(1739/1973, pp. 251–252)

Impressions and Ideas Hume embraced the standard empiricist principles expounded by Locke and Berkeley (and Hobbes). Hume was a psychological atomist, who maintained that all our complex ideas and impressions (of sense or feeling) are composed of discrete simple ideas and impressions:

Simple perceptions or impressions and ideas are such as admit of no distinction nor separation. The complex are the contrary of these, and may be distinguished into parts. Tho' a particular colour, taste, and smell are qualities all united together in this apple, 'tis easy to perceive that they are not the same, but are at least distinguishable from each other.

—(1739/1973, p. 2)

He was also committed to the principle of psychological empiricism. He affirmed that all the simple ideas that compose our complex ideas are derived from impressions of outer sense or inner feeling:

All our simple ideas in their first appearance are deriv'd from simple impressions, which are correspondent to them, and which they exactly represent.

—(1739/1973, p. 4)

Like Locke and Berkeley (and Hobbes), Hume treated ideas as fainter images of impressions (of sense or feeling):

*All the perceptions of the human mind resolve themselves into two distinct kinds, which I shall call *impressions* and *ideas*. The difference betwixt these consists in the degree of force and liveliness with which they strike upon the mind, and make their way into our thought or consciousness. These perceptions, which enter with most force and violence, we name impressions; and under this name I comprehend all our sensations, passions, and emotions, as they make their first appearance in the soul. By ideas I mean the faint images of these in thinking and reasoning.*

—(1739/1973, p. 1)

Like Berkeley, Hume was a meaning empiricist who recognized the critical implications of this principle. He claimed that words that cannot be related to ideas derived from experience are meaningless:

*When we entertain, therefore, any suspicion that a philosophical term is employed without any meaning or idea (as is but too frequent) we need to enquire *from what impression is that supposed idea derived?**

—(1748/1975, p. 22)

But if you cannot point out any such impression, you may be certain you are mistaken, when you imagine you have any such idea.

—(1739/1973, p. 65)

Hume turned his critical eye on our ideas of beauty, causality, material substance, and the self. He declared that we have no ideas of material substance or the self and provided what he thought was the best psychological explanation for our belief in such “fictions.” In the case of our ideas of beauty and causality, he offered a slightly different analysis. He did not deny that we have ideas of beauty or causality, but only that these ideas are derived from sense impressions of the properties of material bodies. Our idea of beauty, for example, is not derived from our sense impression of any property of external material bodies such as the Venus

de Milo, for we have no sense impression of such a property. Instead, our idea of beauty is derived from positive *internal feelings* that are caused by our sensory experience of external material bodies such as the Venus de Milo:

The beauty is not a quality of the circle. . . . It is only the effect which that figure produces upon the mind, whose peculiar fabric of structure renders it susceptible of such sentiments.

—(1751/1975, pp. 291–292)

According to Hume, we mistakenly but quite naturally project this internal idea based upon feelings onto external bodies, as if material bodies themselves had the property of beauty. For Hume, this was just a basic fact about human psychology—we cannot help but project such ideas. Like Francis Hutcheson (1694–1746) before him, Hume treated our idea of beauty as a secondary quality, which is caused by but does not represent any quality of material bodies. Hume famously extended this type of analysis to our idea of causality.

Hume's Fork Hume distinguished between propositions that are rendered true (or false) by relations between ideas, such as “a triangle has three sides” and propositions rendered true (or false) by virtue of corresponding sense impressions or feelings (or lack thereof), such as “all acids are corrosive” or “elation is regularly followed by disappointment.” Propositions such as “a triangle has three sides” are *internally* true, since our idea of a triangle includes the idea of a three-sided figure. By contrast, propositions such as “all acids are corrosive” or “elation is regularly followed by disappointment” are *externally* true, since our ideas of acid and elation do not include the ideas of corrosion or disappointment. We have to discover via sensory or introspective experience whether they are invariably or regularly conjoined.

This distinction between **relation of ideas** and **matters of fact and existence** easily accommodated the propositions of logic and mathematics, “reasoning concerning quantity and number,” whose truth or falsity is determined by internal relations between symbols, and the propositions of empirical science, the products of “experimental reasoning,” whose truth or falsity is determined by independent facts about the world based upon experience (sense impressions or feelings). However, Hume claimed that these two types of propositions are the *only* types of meaningful proposition: If a claim is not identifiable as one of these types, it is meaningless. Thus Hume concluded his *Enquiry Concerning Human Understanding* by proclaiming

When we run over libraries, persuaded of these principles, what havoc must we make? If we take in our hand any volume; of divinity or school metaphysics, for instance; let us ask, *Does it contain any abstract reasoning concerning quantity and number?* No. *Does it*

contain any experimental reasoning concerning matter of fact and existence? No. Commit it then to the flames: for it can contain nothing but sophistry and illusion.

—(1748/1975, p. 165)

Mental Mechanism Hume followed the Newtonian program in psychology by treating simple ideas as the mental atoms or corpuscles out of which all complex ideas are compounded. He took the Newtonian program one step further by employing the principle of the association of ideas as the foundation of mental mechanism. Hume developed universal explanations of mental association in terms of resemblance, contiguity, and cause and effect and conceived of the “uniting principle” of association as the mental analogue of gravity:

Were ideas entirely loose and unconnected, chance alone wou’d join them; and t’is impossible the same simple ideas would fall regularly into complex ones (as they commonly do) without some bond of union among them, some associating quality, by which one idea naturally introduces another. This uniting principle among ideas is not to be consider’d as an inseparable connection; . . . but we are only to regard it as a gentle force, which commonly prevails. . . . The qualities, from which this association arises, and by which the mind is after this manner convey’d from one idea to another, are three, *viz*, RESEMBLANCE, CONTIGUITY in time or place, and CAUSE and EFFECT.

—(1739/1973, pp. 10–11)

Given his commitment to a Newtonian program in psychology, it is probably no accident that Hume employed Newton’s own description of gravity as a “gentle force” in his characterization of the principle of association. It is certainly no accident that Hume characterized the “associating quality” of ideas as a form of attraction, analogous to gravitational attraction:

Here is a kind of ATTRACTION, which in the mental world will be found to have as extraordinary effects as in the natural, and to shew itself in as many and as various forms. Its effects are every where conspicuous; but as to its causes, they are mostly unknown, and must be resolv’d into *original* qualities of human nature, which I pretend not to explain.

—(1739/1973, pp. 12–13)

Even Hume’s modest pretense about his inability to ultimately explain this form of attraction echoes Newton’s avowed ignorance of the ultimate nature of gravity. For example, in his letters to Dr. Bentley, Newton admitted:

The cause of gravity is what I do not pretend to know.

—(Cited in Leon, 1999, pp. 77–78)

Causality as Constant Conjunction Hume's analysis of the idea of (efficient) causality followed the same lines as his analysis of the idea of beauty and exploited the associative principles of resemblance and contiguity. Hume asked his devastating question: From what impression is our idea of causality derived? If one considers a paradigm case of mechanical causation, in which a moving billiard ball collides with a stationary billiard ball and causes it to move, Hume noted that the only common observable features of this sequence are contiguity (togetherness) in space and time, the temporal priority of the motion that we call the cause, and that fact that the two observed motions are "constantly conjoined" in our experience.

Hume recognized that this analysis of the content of our idea of causality is intuitively unsatisfactory. When we believe that the motion of one body causes the motion of another, we do not simply believe that the motions are conjoined in space and time and that the first just happens to be constantly or regularly followed by the second. We believe that the second motion follows the first because the first *produces* or *generates* it: that the force of the collision has the "power" to generate motion in the second ball. As Hume put it, we believe that there is some "necessary connection" between the motion of the first and the second ball: The second ball *must* move, given the prior motion of the first ball. Yet according to Hume, there is no observable feature of causal sequences that corresponds to our idea of power or necessary connection:

It follows that we deceive ourselves, when we imagine we are possest of any idea of this kind, after the manner we commonly understand it. All ideas are deriv'd from, and represent impressions. We never have any impression, that contains any power or efficacy. We never therefore have any idea of power.

—(1739/1973, p. 161)

However, Hume did not really deny that we have any idea of causal power or necessary connection. All he denied is that we have such an idea "as we commonly understand it": that is, as derived from some observed property of material bodies. Instead, in line with his account of our idea of beauty, Hume claimed that our idea of causal power or necessary connection is derived from an internal *feeling*, produced by repeated observations of one event being followed by another. According to Hume, this repetition creates an internal *expectation* of the second event, given an impression or idea of the first. This expectation is based upon an internal feeling only, not upon any observable property of the sequence itself:

For after we have observ'd the resemblance in a sufficient number of instances, we immediately feel a determination of the mind to pass from one object to its usual attendant. The several instances of resembling conjunctions leads us into the notion of power and necessity. These instances are in themselves totally distinct from each other, and have no union but in the mind, which observes them, and

collects their ideas. Necessity, then, is the effect of this observation, and is nothing but an internal impression of the mind, or a determination to carry our thoughts from one to another.

—(1739/1973, p. 165)

Although based upon an internal feeling, we naturally *project* the idea of causality upon external bodies, as if our idea of causal power or necessary connection represented some property of the bodies themselves.

In accord with this analysis, Hume produced two definitions of causality. The first was objective, or “philosophical”:

We may define a CAUSE to be “an object precedent and contiguous to another, and where all the objects resembling the former are plac’d in like relations of precedency and contiguity to those objects, that resemble the latter.”

—(1739/1973, p. 170)

This definition purports to represent the legitimate content of our ascriptions of causality to material bodies in the external world. To say that one thing is a cause of another is just to say that they are contiguous in space and time and constantly conjoined.

Hume’s second definition was psychological, or “natural”:

A CAUSE is an object precedent and contiguous to another, and so united with it, that the idea of the one determines the mind to form the idea of the other, and the impression of the one to form a more lively idea of the other.

—(1739/1973, p. 170)

This definition aimed to provide a psychological account of causal judgment, based upon the associative principles of resemblance and contiguity.

Hume’s account of causal judgment was enormously influential in the development of psychological theories in the 19th and 20th centuries. Although Hume officially recognized three principles of association—namely, resemblance, contiguity, and cause and effect—his analysis of causality reduced the relation of cause and effect to contiguity and similarity with repetition (to the repetition of similar contiguous sequences). The principles of contiguity, similarity, and repetition played a major role in the development of associationist psychology in the 19th century and behaviorist theories of classical and instrumental conditioning in the 20th century.

The Empiricist Conception of Causal Explanation Hume’s objective definition of causality underwrote the later empiricist and positivist conception of causal explanation and scientific laws, including Newton’s laws. In this view, causal

explanations and scientific laws are just descriptions of observational correlation, which are sufficient for prediction and control. Putative explanatory references to “occult” forces or powers are vacuous and redundant. Thus to say that there is a gravitational “force of attraction” between two bodies is just to say that they will move toward each other unless impeded. To say that bodies move toward each other *because* of such a force is just to say that bodies move toward each other because they move toward each other, which is no news to anyone.

One may get the flavor of this type of empiricist and positivist analysis of causal explanation and scientific law by considering a famous example from a Molière play, in which a scholastic doctor offers an explanation of why the ingestion of opium is followed by sleep. Why is the ingestion of opium regularly followed by sleep? Because of the “soporific power” of opium. But wait: What does it mean to claim that opium has a “soporific power”? Well, it means that the ingestion of opium is regularly followed by sleep. Some explanation! But hold on, protests the scholastic doctor: There is more to it than that. Opium has a “soporific power” because of its “dormative nature.” But what does it mean to claim that opium has a “dormative nature”? Well, it means that the ingestion of opium is regularly followed by sleep! According to Hume and later empiricists and positivists, references to “power,” “force,” and “nature” add nothing to causal explanation, and the business of science is just to describe those constant conjunctions of observables that we call scientific laws.

In many respects, Hume’s critique was useful: It put the final nail in the coffin of the ancient (Aristotelian) and medieval notion, still to be found in Descartes, that causal explanations are necessary truths, capable of demonstration through some form of rational intuition—as if by reason alone we could determine that heated water will boil, in the fashion that we determine that the idea of a triangle contains the idea of three sides. For Hume and later generations of philosophers and scientists, causal sequences can only be determined empirically: Nobody can discover what bodies, acids, or human beings can or cannot do except by observation and experiment. Hume placed our knowledge of causality squarely in the realm of empirical “matters of fact and existence” rather than the realm of conceptual “relations of ideas.”

This important contribution deserves due emphasis. However, the virtues of Hume’s analysis have also been exaggerated. It is no doubt true that some explanatory references to “power” and “nature” are vacuous, as in the case of dispositional concepts such as solubility, which can be wholly explicated in terms of observable sequences. To say that something is soluble is just to say that if it is placed in a solvent it will dissolve. Yet this is not obviously true of all theoretical causal explanatory references. Theoretical explanations of the powers of chemical elements in terms of their natures are not mere redescrptions of how they are observed to behave, but descriptions of their underlying composition and structure that explain their observable behavior.

Even references to “gravitational force” need not be explanatorily vacuous, when they are employed to explain motion in terms of the exchange of fundamental particles such as “gravitons” (sometimes held to play the same role in gravitational theory as positrons in the theory of weak nuclear forces). However hesitant Newton may have been about his knowledge of the ultimate nature of gravity, he never doubted the legitimacy of causal explanations of the properties and behavior of material bodies in terms of the motions and interactions of the unobservable corpuscles that compose them (1687/1969, p. xvii).

Although Hume may have been correct in claiming that we have no sensory experience of causal power, it does not follow that causality is nothing but the correlation of observables. Most practicing scientists remain causal realists, who hold that causality is grounded in generative mechanisms, which support qualified conditionals (of the form “if . . . then . . . , unless . . .”) but not descriptions of constant or regular conjunction. Causality cannot be equated with constant or regular conjunction, because although many entities have the causal power to produce certain effects, their action may be prevented or interfered with (Geach, 1975, p. 93). Tin has the power to act as a superconductor when subjected to a low temperature and potential difference, but will not do so in a magnetic field; the tubercle bacillus has the power to induce tuberculosis in humans, but its action can be prevented via inoculation.

For this reason, the frequency of an observed conjunction is no measure of the existence of a causal relation. The sounding of factory hooters regularly follows the sounding of school bells at the end of the day in many localities, but there is no causal relation between them. Conversely, the real power of plutonium rods in nuclear reactors to generate sickness and death in humans is rarely manifested because of lead screening. The frequency with which a particular manifests its causal power is a contingent matter, which depends upon how often the action of a particular is prevented or interfered with, which varies from place to place and over historical time. The incidence of tuberculosis upon exposure to the tubercle bacillus used to be very high in the West, but is now very low because of the development of prophylactics. It is considerably higher in Third World countries with limited vaccination programs and is now on the rise in the West, with the development of bacilli strains resistant to prophylactics.

This is of no small import for a potentially applied science such as psychology, whose practitioners do not merely aim to predict and control behavior, but also hope to be able to intervene to prevent or impede certain forms of behavior, such as aggression, child abuse, suicide, marital breakdown, and interracial conflict, even when causal conditions that promote such behavior are present. To be successful in applied science, it is not sufficient to identify the causal factors responsible for certain types of effects. One also needs to understand the mechanisms underlying causal processes in order to develop effective means of prevention and interference.

One might also have serious doubts about Hume's psychological account of causal judgment. Hume claimed that the strength of our belief in a causal relation is a function of the number of conjunctions we have observed:

As the habit, which produces the association, arises from the frequent conjunction of objects, it must arrive at its perfection by degrees, and must acquire new force from each instance, that falls under the observation. The first instance has little or no force: the second makes some addition to it: the third becomes still more sensible, and 'tis by these slow steps, that our judgment arrives at a full assurance.

—(1739/1973, p. 130)

Yet no amount of repetitions of the school bell being followed by the sounding of the factory hooter, or of the contiguous ringing of two spatially adjacent alarm clocks owned by those who have difficulty rising early, incline us to believe in a causal connection; and children learn that placing their hand in the fire is the cause of their consequent pain the first time around—they do not need to keep sticking their hands in the fire to convince themselves.

David Hartley: The Neurology of Association

David Hartley (1705–1757) extended the Newtonian program of psychology in two fundamental ways. He provided a neurophysiological account of the association of ideas, and he extended the principles of association to encompass behavior. Hartley trained as a minister at the University of Cambridge, but his naturalistic interests led him into medicine. His *Observations on Man: His Frame, His Duty, and His Expectations* were first published in 1749 and ran to six editions (the last published in 1834, almost a century after the original).

Hartley followed earlier empiricists in maintaining that all ideas are derived from experience and tried to explain all mental operations in terms of their association. However, he claimed that temporal contiguity with repetition is sufficient for association and distinguished between “synchronous” and “successive” association:

Any sensations A, B, C, etc., by being associated with one another a sufficient number of times, get a power over the corresponding ideas a, b, c, etc., that any one of the sensations A, when impressed alone, shall be able to excite in the mind b, c, etc., the ideas of the rest. Sensations may be said to be associated together, when their impressions are either made precisely at the same instant of time, or in the contiguous successive instants.

—(1749/1971, p. 65)

Hartley drew inspiration from Newton's theory of gravitation in the development of his neurophysiological theory of association. To explain “action at a distance,” Newton had postulated the ether, which Hartley described as a “very

subtle and elastic fluid . . . diffused through the pores of gross bodies, as well as through the open spaces that are void of gross matter" (1749/1971, p. 13). Vibrations in the ether were held to be the vehicle for the propagation of the effects of gravitation, electricity, magnetism, and "animal sensation and motion" (1749/1971, p. 13). Hartley claimed that sense impressions are instantiated as vibrations in the "white medullary substance" of the brain, generated by "external objects impressed upon the senses" (1749/1971, p. 11). Like earlier empiricists, he treated sensory ideas as weaker or fainter versions of sense impressions: "sensations, by being often repeated, leave certain vestiges, types or images of themselves" (1749/1971, p. 57). He consequently claimed that ideas are instantiated as fainter vibrational traces, or **vibratiuncles** (miniature vibrations) in the brain:

Sensory vibrations, by being often repeated, beget, in the medullary substance of the brain, a disposition to diminutive vibrations, which may also be called vibratiuncles and miniatures, corresponding to themselves respectively.

—(1749/1971, p. 58)

Hartley claimed that association by temporal contiguity is grounded in neural connections between sensations and ideas. For example, repeated conjunctions of sensations of color with those of taste and smell establish neural connections between the vibratiuncles corresponding to their ideas, so that when the vibrations corresponding to the sensations of color are activated via sensory stimulation, they reactivate the weaker vibrations corresponding to the associated ideas of taste and smell (1749/1971, p. 67). The neural connections are established when originally distinct vibrations fuse into a single vibration:

Since the vibrations A and B are impressed together, they must, from the diffusion necessary to vibratory motions, run into one vibration; and consequently, after a number of impressions sufficiently repeated, will leave a trace, or miniature, of themselves, as one vibration, which will recur every now and then, from slight causes. Much later, therefore, may the part b of the compound miniature a + b recur, when the part A of the compound original vibration A + B is impressed.

—(1749/1971, p. 70)

Hartley extended this form of neurophysiologically grounded associationist psychology to include associations of ideas and behavior, via the repeated conjunction of ideas and "motory vibrations":

The motory vibratiuncles will also cohere to ideal ones by association. Common ideas may therefore excite motory vibratiuncles, and consequently be able to contract the muscles.

—(1749/1971, p. 102)

He also claimed that motory vibrations can be associated with each other, which enabled him to offer an explanation of skilled behavior, like playing the piano, in terms of the coordination of motor responses.

Hartley also employed the principle of association to explain learned behavior and habit formation. He distinguished between voluntary behavior, brought under the control of ideas (or “affections”) through association, and automatic (involuntary) behavior, the product of sensory-motor associations unmediated by conscious thought. He treated many automatic behaviors as reflexive behaviors based upon vibratory connections in the spinal cord rather than the brain, although he also noted how certain skilled behavior that is originally consciously controlled, such as serving a tennis ball, becomes “secondary automatic” or habitual with practice. Conversely, he claimed that through ideomotor association we are able to gain voluntary control over some reflexive behavior, such as “swallowing, breathing, coughing and expelling the urine and faeces” (1749/1971, p. 108).

In developing his account of how originally reflexive behavior can come under the control of ideas via repeated temporal contiguity, Hartley provided an early account of classical conditioning. He noted how a child’s reflexive muscular response to a toy can become conditioned to the mere sight of a toy:

The fingers of young children bend upon almost every impression which is made upon the palm of the hand, thus performing the action of grasping, in the original automatic manner. After a sufficient repetition of the motory vibrations which concur with this action, their vibratiuncles are generated, and associated strongly with other vibrations and vibratiuncles, the most common of which, I suppose, are those excited by the sight of a favorite play thing which the child uses to grasp, and hold in its hand. He ought, therefore, according to the doctrine of association, to perform and repeat the action of grasping, upon having such a play thing presented to his sight. But it is a known fact, that children do this.

—(1749/1971, pp. 104–105)

Hartley also developed an account of motivation through association that later came to be known as instrumental (or operant) conditioning. He identified pleasure and pain with moderate and excessive vibration respectively, pain being nothing more than pleasure “carried beyond its due limit,” and held that all ideas and behavior are “attended to some degree by pleasure and pain” (1749/1971, p. 9). He followed the Reverend John Gay (1699–1745), who claimed that human behavior is regulated by associations of behavior with pleasure and pain, including imaginary pleasure and pain. Given that humans seek pleasure and avoid pain, they tend to pursue behavior that has come to be associated with pleasure and avoid behavior that has come to be associated with pain. Gay’s views were published anonymously in 1731 in an essay titled “Dissertation on the Fundamental Principle of Virtue.” Hartley claimed that Gay’s “Dissertation” was the stimulus for

his own work, and he reaffirmed Gay's contention that the association of behavior with pleasure and pain is the fundamental principle of morality, a view later developed as utilitarian theory by Jeremy Bentham (1748–1832) and John Stuart Mill (1806–1873).

Hartley differed from earlier empiricists in one critical respect. Although he followed them in treating complex ideas as compounded out of simpler atomistic elements, he maintained that the formation of complex ideas is more akin to chemical fusion than mechanical association. According to Hartley, the simple sensory ideas that compose complex ideas fuse into unitary ideas (as associated vibrations fuse into single vibrations), which bear little relation to the simple ideas from which they are generated:

If the number of simple ideas which compose the complex one be very great, it may happen, that the complex idea shall not appear to bear any relation to its compounding parts, nor to the external senses upon which the original sensations, which gave birth to the corresponding ideas, were impressed.

—(1749/1971, p. 402)

Consequently, in our perception or thought of an apple, for example, we are not aware of the sensational elements of color, texture, smell, and taste from which it is generated.

Hartley's *Observations* had little immediate impact (partly because of his rather turgid prose), although it was an important influence on James Mill (1773–1836) and his son John Stuart Mill, who went on to develop their own versions of associationist psychology and utilitarian theory. It had a greater impact when Hartley's views were championed by the English chemist and political radical Joseph Priestley (1733–1844), who produced an edited reprint of Hartley's *Observations* in 1775 titled *Hartley's Theory of the Human Mind, on the Principle of the Association of Ideas*. Priestley's advocacy of a materialist psychology based upon the principles of association attracted the same degree of odium as La Mettrie's *Man Machine*, and Priestley was driven out of England as La Mettrie had formerly been driven out of France and Holland. This was somewhat ironic, since Priestley had cut most of Hartley's (already outdated) discussion of neural vibrations from the reprint. It was also unfair to Hartley, who affirmed the existence of an immaterial soul and saw his own work as an exercise in natural theology, demonstrating the benevolence of God through the study of nature, including human nature.

Yet the distance between Hartley and La Mettrie was not great. Hartley also affirmed the strong continuity between human and animal psychology and behavior that La Mettrie had championed (Boakes, 1984). Although Hartley was committed to the existence of an immaterial soul that survives the destruction of the material body, it was an impoverished sort of thing, stripped of the essential

Cartesian property of thought. Hartley claimed that since the soul depends upon the brain and body for all its cognitive and affective operations, it is “reduced to a state of inactivity by the decomposition of the gross body” (1749/1971, p. 402) and remains in a “dormant” state until reincarnated at the Resurrection.

Priestley’s advocacy of Hartley’s theories had one significant consequence. Erasmus Darwin (1731–1802), the grandfather of Charles Darwin and a friend of Priestley’s, generalized Hartley’s neurophysiologically based associative account of how habits are acquired during the lifetime of individuals to provide an account of the maintenance of habits over generations of a species and developed an early account of the evolution of species in *Zoonomia: Or the Laws of Organic Life* (1794–1796). Darwin was a physician and, like Priestley, an uncompromising materialist. He stressed that associations are wholly determined by neurophysiology and dismissed appeals to the action of immaterial souls as “ghost stories.”

Sensationalists and Ideologues in France

Sensationalists such as Étienne Bonnot de Condillac (1715–1780) and Claude Helvetius (1715–1771) developed the empiricist psychology of Locke, Berkeley, and Hume in France. Condillac was a great admirer of Locke: He translated Locke’s *Essay Concerning Human Understanding* into French and developed his theories in *Essay on the Origin of Human Knowledge: A Supplement to Mr. Locke’s Essay on the Human Understanding* (1746) and *Treatise on the Sensations* (1754). Condillac provided a stripped-down version of Locke’s psychology: He held that all our ideas and mental faculties (which even Locke had supposed were innate) could be accounted for in terms of sensation, or sensation “transformed.” He claimed that mental faculties are products of the intrinsic pleasure and pain of sensation, which stimulates the development of attention, comparison, imagination, memory, and reflection (based upon the employment of words as signs for ideas derived from sensation). Condillac famously illustrated these claims by imagining a statue possessing the single modality of smell, which he held to be the simplest form of sensation. This enabled him to claim that mental capacities are not grounded in the integration of different sensory modalities, such as an Aristotelian “common sense,” and to distinguish the distinctive contribution of the different sensory modalities. For example, Condillac claimed that ideas of external objects are derived from the sense of touch.

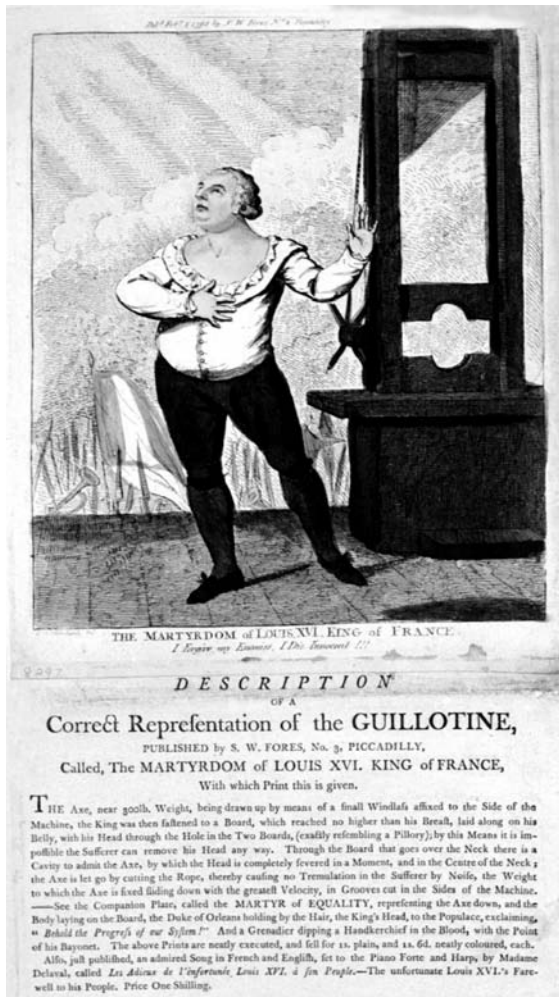
Many rejected Condillac’s theory as materialistic, notably Victor Cousin (1792–1867), the French critic of Locke, although it was more developmental than reductionist. Condillac was a former Catholic priest who railed against religious dogma, but affirmed the existence of the immaterial soul. He claimed that sensation enables the soul to attain knowledge, a view also promoted by the Swiss religious apologist Charles Bonnet (1720–1793). Like Locke, Condillac’s interest in

the development of ideas led him to focus on the educational implications of his theories, which he developed in *Logic* in 1780.

Claude Helvetius also promoted the developmental, educational, and political implications of psychological empiricism. He published *Essays on the Mind* in 1758; his *Treatise on Man, His Intellectual Faculties and His Education* was published posthumously in 1772. A radical and optimistic environmentalist, Helvetius reasoned that if all knowledge comes from experience, and behavior is motivated through association with consequent pleasure or pain, then virtually anything could be inculcated through the social manipulation of experience through education and legislation: science, morality, even genius. According to Helvetius, the primary impediment to progress through education and legislation is religious dogma. Yet like Hobbes, he maintained that the unbridled pursuit of pleasure and avoidance of pain would lead to selfish and destructive behavior. This could be constrained only through social approbation and directed education and laws that ensured, through the sanction of punishment, that the interests of individuals included the interests of others. Consequently Helvetius claimed that society, education, and law are the foundation of human character and virtue.

Although many criticized their theories, notably the clergy (the faculty of the Sorbonne condemned and burned Helvetius's *Essays on the Mind*), Condillac and Helvetius exerted a powerful influence in the decades before and immediately after the French Revolution. Their theories were embraced by the contributors to the multi-volumed *Encyclopédie*, the Bible of the French Enlightenment that aimed to provide a comprehensive treatment of the various branches of human knowledge. They were also adopted by *idéologues* such as Antoine-Louis-Claude, comte Destutt de Tracy (1754–1836) and Pierre Jean George Cabanis (1757–1808), who played a significant role in reshaping French higher education and the legal and medical profession in the early days of the French Republic. Both served as supervisors of general education in the period between the Terror and the rise of Napoleon, who promptly banned their work. Their vision of a socially applied psychology provided a powerful inspiration for later scientific psychologists and social and political theorists in Europe and America, and their optimistic philosophy exerted an immediate influence on liberal theorists such as Thomas Jefferson (1743–1826), the author of the Declaration of Independence and the third American president (Robinson, 2003).

De Tracy, who claimed that both the development and application of human knowledge depends upon ideas, coined the term *ideology*. De Tracy held that “custom” is the source of human misery, but also the best hope for human progress through social and educational redirection. Cabanis provided a neurophysiological foundation for the psychology of Condillac and Helvetius in *Studies on the Physical and Moral Nature of Man* in 1799. A physician and admirer of La Mettrie, Cabanis held that the brain secretes thought as the stomach secretes gastric juices,



The guillotine: the empirical basis of Cabanis's theory that the brain is the seat of consciousness and that many sensory-motor reflexes are governed by the spinal cord.

which naturally drew down familiar charges of materialism and threats to morality and religion. Based upon his study of decapitated victims of the guillotine, Cabanis claimed that while the brain serves the central ego, the seat of consciousness, will, and rationality, many sensory-motor reflexes are governed by the spinal cord (a claim that had been advanced earlier by Hales and Whytt). He also claimed that the study of abnormality and sociality, in addition to development and physiology, is central to a proper theoretical understanding of human psychology and behavior—a programmatic statement, but one that clearly anticipated the later development of scientific psychology.

CRITICAL RESPONSES TO NEWTONIAN PSYCHOLOGY

Not all theorists embraced the notion of a scientific psychology based upon the principles of Newtonian science, especially as interpreted by British empiricists and French sensationalists and ideologues. Scottish realist philosophers such as Thomas Reid (1710–1796) and Dugald Stewart (1753–1828) developed a form of “common sense” psychology based upon the direct perception of material bodies and their properties and maintained that scientific, moral, and religious knowledge is grounded in innate powers or faculties bestowed upon humanity

by a benevolent God. German rationalist philosophers such as Gottfried Leibniz (1646–1716) and Immanuel Kant (1724–1804) also claimed that certain ideas and forms of knowledge are innate and questioned the empiricist view that perception is based upon the association of sensory elements. More radical critics such as Giambattista Vico (1668–1744) rejected the common assumption that human sciences such as psychology should be modeled upon natural sciences such as physics and questioned whether psychology should be based upon Newtonian principles such as the universality of explanation and ontological invariance. Romantics such

as Jean-Jacques Rousseau (1712–1778), Johann Wolfgang von Goethe (1749–1832), Arthur Schopenhauer (1788–1860), Friedrich Nietzsche (1844–1900), and Georg Wilhelm Hegel (1770–1831) rejected the general Enlightenment attempt to confine human spontaneity and creativity within the limits of reason and science.

Realism and Common Sense

Thomas Reid mounted one of the most sustained critiques of psychological atomism and associationist psychology, since he believed that such doctrines promoted a fatal combination of materialism, determinism, skepticism, and atheism. Reid was born and raised in Aberdeen, Scotland, where he served a short time as a Presbyterian minister before taking up a position as professor of moral philosophy at King's College, Aberdeen, in 1751. He became professor of moral philosophy at the University of Glasgow in 1764. Reid established what came to be known as the Scottish school of "common sense" psychology with the publication of *Enquiry into The Human Mind on the Principles of Common Sense* in 1764, followed by *Essays on the Intellectual Powers of Man* in 1785, and *Essays on the Active Powers of Man* in 1788.

Reid denied the fundamental tenets of psychological atomism endorsed by Locke, Berkeley, and Hume and recognized the fundamental problem of any psychology based upon the Lockean "way of ideas": Our perception and knowledge of physical bodies with shape, size, and motion cannot be accounted for in terms of the mere aggregation or association of atomistic sense-impressions. This had led Locke and Hume to treat perception and knowledge of physical bodies as a doubtful inference, and Berkeley to embrace idealism by claiming that our ideas about physical bodies refer to nothing but the association of atomistic sense-impressions. In contrast, Reid advocated a form of **direct realism**: He claimed that we directly perceive physical bodies and their properties without the mediation of atomistic sense impressions.

Reid distinguished between physical stimulation, sensation, and perception: between, for example, the physical stimulation of the retina, the sensation of color, and the perception of an apple. He maintained that sensation cannot be explained in terms of physical stimuli and that the intentional perception of physical bodies such as apples cannot be explained in terms of nonintentional states such as sensations of color, smell, and taste. Consequently, Reid vehemently rejected the form of associationist psychology grounded in neurophysiology propounded by Hartley, whose *Observations* Reid described as a "fallacious tract," and he denied the possibility of a science of psychology modeled upon the mechanistic program of Galileo and Newton.

Reid claimed that the active power to directly perceive physical bodies and their properties, such as their shape, size, distance, and motion, is part of the

constitutional “common sense” of humanity. These are the set of common powers provided by a benevolent God, which are ideally suited to their purpose:

When I perceive a tree before me, my faculty of seeing gives me not only a notion or simple apprehension of a tree, but a belief of its existence, and of its figure, distance and magnitude; and this judgment or belief is not got by comparing ideas, it is included in the very nature of the perception. . . .

Such original and natural judgments are, therefore, a part of the furniture which Nature hath given to the human understanding. They are the inspiration of the Almighty, no less than our notions or simple apprehensions. They serve to direct us in the common affairs of life, where our reasoning faculty would leave us in the dark. They are part of our constitution; and all the discoveries of our reason are grounded upon them. They make up what is called *the common sense of mankind*.

—(1764/1975, p. 188)

According to Reid, God ensures that perception generally accords with physical reality. There is no need to explain perception in mechanistic, atomistic, and associative terms and no possibility of doing so. Reid described a range of innate faculties that supposedly ground our mental powers, including our moral and religious sensibilities, whose reliability is guaranteed by their divine endowment. In this fashion he defended the common judgments of mankind against materialism, determinism, skepticism, and atheism and developed a purely descriptive psychology that documented innate human powers and faculties. His claim about the common cognitive, moral, and religious faculties of humankind became the foundational tenet of Scottish **common sense psychology**.

Reid’s students and disciples, Dugald Stewart, Thomas Brown (1778–1820), and William Hamilton (1788–1856), reaffirmed his theory of innate faculties and denial of a mechanistic science of psychology, but also continued to develop associationist psychology (albeit shorn of its mechanistic and materialist trappings). Stewart, one of Reid’s students, and later professor of moral philosophy at the University of Edinburgh, promoted associationist psychology in *Elements of the Philosophy of the Human Mind* (1792), while insisting that association is not susceptible to scientific analysis. Thomas Brown, who succeeded Stewart at Edinburgh (and was Stewart’s own student), abandoned Reid’s direct realism, because he felt it was too materialistic, and employed Hume’s analysis of causation as constant conjunction to critique the materialist psychology of Darwin’s *Zoonomia*. He maintained that Darwin merely described the correlation between mind and matter, not their causal relation. Brown developed a number of secondary laws of association, or “suggestion,” and postulated a “muscle sense” to account for our perception of the externality of material objects, in terms of associations linked to feelings of resistance. William Hamilton, professor of logic and metaphysics at

Edinburgh, reaffirmed the directness of perception, albeit “conditioned” by the nature of the perceiving subject, and continued to deny the relevance of mechanistic science to psychology.

Common sense psychology was enormously influential in Europe and America. It was developed in France through the energies of the Academician Victor Cousin, who ensured that common sense psychology displaced the theories of the sensationalists and idéologues in French higher education. It was especially influential in America, where it was carried by generations of Presbyterian ministers trained in the common sense psychology of the Scottish universities, many of whom became presidents of American universities. John Witherspoon (1723–1794), the president of the College of New Jersey, later Princeton University, introduced Scottish common sense psychology in his lectures on moral philosophy. Stewart’s *Elements of the Philosophy of the Human Mind* (1792), along with Brown’s collected *Lectures on the Philosophy of the Human Mind* (1820) and James Abercrombie’s *Inquiries Concerning the Intellectual Powers and the Investigation of Truth* (1830), were standard textbooks at American colleges and universities in the 19th century.

Rationalist Reaction

Another form of critical reaction came from rationalist philosophers in Germany, who claimed that some ideas and forms of knowledge are innate and that the mind plays a much more active role in perception and cognition than empiricists recognized. They also maintained that many mental states and processes are unconscious and rejected the traditional treatment of ideas or concepts as images derived from sense impressions.

Leibniz and Apperception Gottfried Wilhelm Leibniz was a polymath who made major contributions to logic, mathematics, jurisprudence, and history, in addition to philosophy and psychology. He developed the differential and integral calculus independently of Newton and possibly before him. He also developed the notion of a universal logical language that forms the theoretical basis of modern computing devices. Leibniz entered the University of Leipzig at age 15 and presented his thesis for the degree of doctor of law at age 20. He traveled widely in Europe, where he met or corresponded with most of the major figures of his day. He became librarian to the Duke of Brunswick at Hanover, a position he held until his death.

Leibniz’s first work and major psychological thesis was *New Essays on the Understanding*, a response to Locke’s essay completed in 1704 but not published until 1765. This was because Locke died in 1704, and Leibniz delayed publication out of respect. Whereas Condillac had complained that Locke granted too many innate capacities to the human mind, Leibniz complained that he granted too few. In particular, Leibniz complained that Locke had neglected our abstract knowledge

of mathematics and science, based upon our innate ideas of number, space, time, substance, and causality. In a prescient metaphor, he conceived of these ideas as analogous to the outline of a statue of Hercules marked out in the veins of a block of marble, both of which require development to become manifest:

Hercules would be innate in it, in a way, even though labor would be required to expose the veins and to polish them into clarity, removing everything that prevents their being seen. This is how ideas and truths are innate in us—as inclinations, dispositions, tendencies, or natural potentialities.

—(1765/1981, p. 46)

Leibniz's theoretical system is strange and intricate and distributed over a number of works. However, two aspects of his complicated system came to play a major role in the later development of psychology. Leibniz maintained that there are sensory impressions that are perceptually registered but so faintly that they do not enter consciousness, which he called **petites perceptions**. Sometimes the combined intensity of these *petites perceptions* is sufficient to generate perceptual awareness, or **apperception**. As an illustrative example, he noted how the perceived sound of a crashing ocean wave is composed of individually indistinguishable sounds produced by individual droplets of water. According to Leibniz, apperception is not a product of the passive aggregation of sensory elements, but of the active organization of sensory elements into a unified perceptual whole—an account that exerted a major influence on later theorists, notably the Gestalt psychologists. Leibniz also introduced the notion of a **sensory threshold**, below which sensory impressions do not register in consciousness (such as the sound produced by an individual droplet of water). This notion, anathema to empiricists such as Locke, Berkeley, and Hume, had a powerful influence on later theorists such as Johann Friedrich Herbart (1776–1841) and Moritz Drobisch (1802–1896) and became a central feature of the psychophysics of Gustav Fechner (1801–1887).

Kant and the Categories Immanuel Kant was one of the greatest philosophers of the modern period. He was born in Königsberg, Prussia, and educated at the University of Königsberg, where he taught until his retirement at the age of 73. Kant's life was a paradigm of mundane order. He never married and left Königsberg only once (to visit a friend in a town 40 miles away): the townsfolk were said to set their watches according to the legendary punctuality of his daily walks. His early works were devoted primarily to physics and astronomy: He predicted the existence of the planet Neptune, later discovered by Herschel. However, in his middle age he developed what came to be known as his “critical philosophy,” for which he became so famous that he had to keep changing restaurants to avoid the crowds of admirers that came to watch him eat lunch.

Kant's theories cannot be easily fitted into the traditional categories of "rationalism" or "empiricism," and his mature critical philosophy was, by his own account, an attempt to create "a Copernican revolution in thought." In the *Critique of Pure Reason*, published in 1781, Kant argued that the form (although not the specific content) of our knowledge of the external world is based upon a number of innate principles, or **categories**, of thought. For Kant, space and time represent the innate form of our sensory experience, which the mind actively organizes—through apperception—to form empirical concepts regulated by innate categories of substance, causality, unity, plurality, necessity, possibility, and the like.

Kant recognized the central problem of empiricist psychological accounts of our perception and knowledge of material bodies causally related in space and time. Our ideas or concepts of material substance and causality, for example, do not appear to be derived from sensory experience or constructed from atomistic ideas derived from sensory experience. The idea of material substance, for example, does not appear to be derived from the mere association of sensory impressions or ideas: Rather, it purports to represent an enduring particular with sensible properties. For this reason, according to Kant, empiricist treatments of our concepts of material substance and causality in terms of the "constancy and coherence" and "constant conjunction" of sensory impressions are hopelessly inadequate. With some justice Kant credited Hume as having woken him from his "dogmatic slumbers."

Since Kant held that the forms of sensibility and categorical concepts structure the form of our knowledge, he claimed that we have **synthetic a priori knowledge** of the fundamental principles of Newtonian physical science: We have forms of knowledge of the natural world that are not based upon experience. For example, Kant claimed that we can know a priori that throughout all change the quantum in nature remains constant, that every event has a cause, and that for every action there is an equal and opposite reaction. Although he maintained that such principles ground scientific disciplines, he insisted that particular causal laws can be determined only by observation and experiment. He also maintained that synthetic a priori knowledge is only knowledge of things in the external world *as they appear to our senses*, bound by the spatial and temporal forms of our sensibility, and not knowledge of things *as they are in themselves*. For this reason, Kant characterized his critical philosophy as **transcendental idealism**—transcendental because it described the conditions of the possibility of experience.

Kant was one the few theorists of the period to distinguish between sense perception and cognition, while stressing the necessary contribution of both to our knowledge of the world. As he famously put it, "thoughts without content are empty; intuitions [sense impressions] without concepts are blind" (1781/1973, p. 92). In contrast to Descartes, Locke, Berkeley, Hume, and Hartley, Kant denied that our empirical concepts of apples, trees, and tables are images of our sensory experience of them. Rather, our empirical concepts are cognitive schema for

objects in space and time, which enable us both to re-identify instances of material substances such as apples, trees, and tables, and to form images of them.

Kant is famous for his supposed denial of the possibility of a science of psychology. In the *Metaphysical Foundations of Natural Science* he claimed that “empirical psychology [must] be removed from the rank of a natural science so called” (1783/1891, p. 8), because it could not be quantified and because self-observation would alter the objects of any putative science of “inner sense.” Kant did deny the possibility of a scientific psychology based upon the introspective analysis of the association of sensations and ideas. This was because he maintained that such a psychology would be restricted to the description of correlation in a single temporal dimension and would not constitute a scientific psychological analogue of dynamical physics, which requires the four dimensions of space and time (motion being defined as change of position in three-dimensional space over time).

However, it is a misrepresentation to characterize Kant as having denied the possibility of a science of psychology altogether. He acknowledged that it is possible to provide quantitative measures of the intensity of sensation (in the *Anticipations of Perception* section of the *Critique of Pure Reason*), as later developed in Fechner’s psychophysics (see also Sturm, 2006). He also recognized the potential of social and developmental psychology, which he detailed in his *Anthropology From a Pragmatic Point of View* (1797/1996). This work, which contains a marvelous warning about how introspective overindulgence can lead to madness, was representative of a radically different conception of psychological science.

Something Completely Different

From the time of the scientific revolution to the present day, it has been common to characterize psychology and other human sciences as the inferior relatives of the natural sciences, which they can at best approximate. Proponents of psychological science from Hume to Watson have argued that psychology can become an objective scientific discipline only by emulating the methods, explanatory modes, and principles of the natural sciences and that forms of psychological knowledge can at best merely approximate those of natural scientific knowledge.

Vico and Human Science In stark contrast, Giambattista Vico argued that the objects of psychology and other human sciences are better known than the objects of natural science. Born in Naples, Vico taught rhetoric at the University of Naples. In the *New Science* (1725, reprinted in 1730 and 1744), he followed Aristotle in claiming that the governing principles of any entity or process are best known to its creator. Consequently, Vico argued that we know the forms of thought, emotion, and behavior that are the created products of our social

being better than we know the world of nature, which can be known only by God: “since God made it, He alone knows.” We have direct knowledge of our own internal psychological states through introspection and empathetic insight into the psychologies of other peoples, whereas we have only indirect knowledge of the fundamental properties of material substances through our theoretical representations or models of them.

Vico also rejected standard Newtonian assumptions of invariance and universality. He believed that many of the fundamental principles of human psychology and behavior are developed products of human culture and history and possibly as diverse as them. He recognized the problem this created for our understanding of the psychology of persons living in other cultures and at different historical periods. How can we understand their “web of belief” in terms of our own culturally and historically specific web of belief? Vico suggested we could make some inroads to understanding by considering the varied attitudes and practices of different communities relating to what appear to be the few cultural and historical invariants of human life, namely birth, sex, and death (1725/1984, pp. 332–333).

Although neglected in his own day, Vico’s distinctive contribution marked the beginning of a tradition of social and developmental psychology that took seriously the possibility that the explanatory principles of human psychology and behavior vary cross-culturally and transhistorically. Vico’s basic principles were restated by Johann Gottfried Herder (1744–1803) in his *Ideas Towards a Philosophy of History* (1784–1791). They were represented in Kant’s *Anthropology From a Pragmatic Point of View* (1797) and John Stuart Mill’s ethological “science of character,” which Mill maintained was a necessary supplement to associationist psychology (Mill, 1843/1973–1974). They were also developed in Wundt’s *Volkerpsychologie* (1900–1920), which Wundt treated as a complement to experimental psychology.

These theorists maintained that a distinctive feature of psychology and other human sciences is an understanding of the *meaning* of human thought and behavior. Later critics opposed to 19th-century forms of psychology grounded in empiricist and positivist conceptions of science developed this feature as the basis of a principled distinction between natural and human sciences. According to the German philosopher and historian Wilhelm Dilthey (1833–1911), the goal of the natural sciences (*Naturwissenschaften*) is mechanistic causal explanation (*erklären*), whereas the goal of human sciences (*Geisteswissenschaften*) is interpretive understanding (*verstehen*).

Romanticism

These reactions to a Newtonian science of psychology were paralleled by a repudiation of the Enlightenment ideals of reason and science by the **romantic** movement, represented by theorists such as Jean-Jacques Rousseau, Johann Wolfgang von Goethe, Arthur Schopenhauer, and Friedrich Nietzsche. Rousseau argued,

first in the *Prize Essay for the Dijon Academy* in 1749 and later in *The Social Contract* (1762/1997), that human nature has been corrupted by reason and science. According to Rousseau, human beings in their natural state are “noble savages.” They are naturally inclined to develop into free, fulfilled social beings, but these spontaneous impulses are corrupted by civilization. Consequently the ideal form of education is one in which the child’s natural inclinations are encouraged and nurtured, a form of education Rousseau detailed (albeit in an idealized fashion) in *Emile* (1762/1979).

Rousseau commended the spontaneity of emotion over the sterility and artificiality of reason and science. Goethe, while not opposed to science, claimed that there are limits to the rational scientific approach to human nature. He suggested that certain human attributes, such as creativity and the capacity for self-transformation through a passionate approach to life, transcend scientific understanding. Schopenhauer focused on the irrational aspects of the human will, which he maintained is driven by fundamental needs that are continually frustrated. Nietzsche hymned the irrational, passionate, and impulsive side of human nature. Freud later developed his account of the conflict between the emotional and impulsive, and the rational and repressive, aspects of human nature.

Although romantics repudiated Enlightenment faith in reason and science, they shared the Enlightenment vision of human history as progressive and purposive, albeit conceived as a form of spiritual journey—thus the common theme of quests, journeys, and pilgrimages in romantic art. This conception reached its apotheosis in the psychological, social, and political theory of the German philosopher Georg Wilhelm Hegel (1770–1831). Hegel represented human history as a progress toward freedom, in which the human mind develops through all possible forms of experience to attain knowledge of the self and the world as it is in itself, or the Absolute. Thus Hegel’s philosophy is often characterized as Absolute idealism.

In discussing the possible forms of experience, mentality, and consciousness, Hegel recognized the social dimensions of mind, the orientation of the thought and emotion of individuals to the represented thought and emotion of other individuals in a social group. He claimed that an individual’s own feelings of self-respect are grounded in the represented respect of others. Unfortunately, Hegel elevated this account of the intersubjectivity of social thought and emotion into an ontological doctrine about the **social mind** (or spirit) of social groups, states, and nations, conceived of as an emergent form of mentality irreducible to the mentality of the members of social groups, states, and nations. Hegel’s mystical conception of the state had disastrous consequences for later German history, by promoting unquestioning obedience to the authority of the state. His notion of social and political progress in history as a form of dialectical resolution of “contradictions” within historical epochs was the inspiration for the dialectical materialism of Karl Marx (1818–1883), the founder of revolutionary communism.

TOWARD A SCIENCE OF PSYCHOLOGY

These reactions to Newtonian psychology impeded but did not prevent the development of scientific psychology, which was instituted as an autonomous academic discipline in Germany and America at the end of the 19th century. Nonetheless, scientific psychology did not develop directly from David Hartley's neurophysiologically grounded form of associationist psychology, which was largely ignored in his own time and condemned when Priestley promoted it as materialist psychology. Although associationist psychology continued to develop in Britain and France, it was taught in British, French, and American universities only as an aspect of Scottish common sense psychology, which repudiated the materialistic and mechanistic basis of Hartley's theory.

The 19th century saw great advances in neuroanatomy, physiology, and evolutionary theory, which powerfully shaped the development of scientific psychology in the late 19th and early 20th centuries. Yet the emergence of institutionalized academic psychology in Germany and America in the 1880s was scarcely a triumph of materialism. Like associationist psychologists, many 19th-century physiologists and evolutionary theorists were careful to avoid or downplay questions about the material basis of mind. Some remained committed to interactive dualism, while others avowed a neutral parallelism and refused to speculate about the nature of the correlation between mental and material brain states.

DISCUSSION QUESTIONS

1. Is it reasonable to anticipate that some day psychologists will develop a universal theory that will unify the various subtheories and branches of psychology, in the fashion that Newton's gravitational theory unified the various laws of physical motion?
2. Why do you think that Locke was so convinced that consciousness is inseparable from thought? Can you be happy or miserable without being conscious that you are?
3. According to Berkeley, all concepts or ideas are derived from experience, but we can neither see nor touch distance. Is there any way to account for our ability to judge distance without having to postulate innate concepts of distance?
4. Try Hume's experiment of trying to "catch yourself" in introspection? Do you encounter nothing but perceptions, memories, feelings, and the like?
5. Do you agree with Condillac that it would be possible for a cognitive being to have only one sensory modality? Would this be sufficient (in principle) for the development of all cognitive faculties?

6. In what sense may we be said to collectively make our own psychology?
Does Vico's account apply to all areas of psychology? To any areas?

GLOSSARY

apperception Term employed by Leibniz and Kant to describe the active organization of sensory elements in perception.

categories According to Kant, the innate concepts that organize and structure perception and knowledge.

Common sense psychology The form of psychology based upon the direct realism of Thomas Reid, according to which perception and knowledge are grounded in common and innate powers or faculties.

corpuscularian hypothesis The hypothesis that the properties and behavior of complex material bodies can be explained in terms of the properties and behavior of the corpuscles (or atoms) that compose them.

direct realism The view that we directly perceive physical bodies and their properties, without the mediation of atomistic sense impressions.

Encyclopédie A set of books produced in France in the 18th century under the direction of Denis Diderot that aimed to provide a comprehensive treatment of the various branches of human knowledge.

empiricist/positivist conception of scientific explanation The conception of scientific explanation as nothing more than the description of observational correlation.

epistemological empiricism The view that all knowledge derives from experience.

Enlightenment The period of European thought in the 17th and 18th centuries in which confidence in reason and experience came to displace faith in religion and traditional authority.

idealism The view that only immaterial minds and their ideas exist.

idéologues French empiricists dedicated to human progress through the application of psychology to social reform and education.

matters of fact and existence According to Hume, propositions rendered true or false by experience.

petites perceptions Sensory elements that register in perception but that are too faint to enter consciousness.

psychological atomism The view that mental states can be individuated independently of each other, and that complex ideas or concepts are compounded out of distinct simple ideas or concepts.

- relations of ideas** According to Hume, propositions rendered true or false by conceptual relations between ideas.
- romanticism** Eighteenth-century movement that repudiated the Enlightenment ideals of reason and science and celebrated human emotionality, spontaneity, and creativity.
- sensationalism** The French versions of empiricist psychology developed by Condillac and Helvetius.
- sensory threshold** The level below which sensory impressions do not register in consciousness.
- social mind** Conception of social thought and emotion as emergent forms of mentality of social groups, states, and nations that are irreducible to the mentality of their members.
- synthetic a priori knowledge** According to Kant, knowledge about the natural world that is not based upon experience.
- theoretical unification** View that science progresses through the theoretical unification of independently established scientific laws.
- transcendental idealism** Kant's account of synthetic a priori knowledge, based upon his analysis of the conditions of the possibility of experience.
- vibratiuncles** According to Hartley, neural vibrations that form the material basis of ideas.

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Physiology and Psychology

THE 19TH CENTURY WAS A TIME OF GREAT CHANGE IN EUROPE AND America. Agricultural reforms ensured a steady food supply, and improvements in public hygiene decreased fatalities due to contagious diseases such as cholera. The population of Europe increased from about 140 to 420 million people between 1750 and 1900, with many congregated in the new urban centers. The dramatic expansion of industry led to a general increase in wealth, although the insecurities of the capitalist state (with periods of boom followed by periods of economic downturn) led some to question a system in which most of the wealth was owned by a privileged few and to look to alternative political systems such as socialism and communism. New developments in transportation and communication saw the spread of modern road networks, railways, canals, ocean lines, and telegraph and postal systems (Jansz, 2004).

The 19th century witnessed the growth and increasing political strength of the middle class, whose long struggle to attain voting rights eventually bore fruit, although throughout most of the 19th century real political control remained in the hands of the conservative aristocracy. In the reactionary period following the Napoleonic wars in Europe, which ended with Napoleon's defeat at the battle of Waterloo in 1815, naturalistic approaches to psychology were repressed through censorship and the secret police. Nobody who promoted such views could hold a professorship in Europe and America in the early half of the century, and in the years immediately following 1815, advocacy of such views was punishable by imprisonment in some parts of Europe (Reed, 1997).

Joseph Priestley (1733–1804) was hounded out of Britain for his promotion of Hartley's associationist psychology as materialist psychology. Erasmus Darwin (1731–1802), the grandfather of Charles Darwin (1809–1882), who developed an early naturalistic evolutionary theory in *Zoonomia* (1794–1796), found his work suppressed. One of Darwin's followers, the British surgeon William Lawrence (1783–1867), published his theory that insanity is a neurophysiological disorder in *Lectures on Physiology, Zoology and the Natural History of Man* (1819). The medical establishment forced him to withdraw his book, and he lost his lectureship at the Royal College of Surgeons (Reed, 1997).